

Table 13.7 Trading Simulation Performance Measures

Performance Measure	Description
Number of Periods Daily Returns Rise	$NPR = \sum_{t=1}^N Q_t \quad (13.57)$ where $Q_t = 1$ if $y_t > 0$ else $Q_t = 0$
Number of Periods Daily Returns Fall	$NPF = \sum_{t=1}^N S_t \quad (13.58)$ where $S_t = 1$ if $y_t < 0$ else $S_t = 0$
Number of Winning Up Periods	$NWU = \sum_{t=1}^N B_t \quad (13.59)$ where $B_t = 1$ if $R_t > 0$ and $y_t > 0$ else $B_t = 0$
Number of Winning Down Periods	$NWD = \sum_{t=1}^N E_t \quad (13.60)$ where $E_t = 1$ if $R_t > 0$ and $y_t < 0$ else $E_t = 0$
Winning Up Periods (%)	$WUP = 100 \cdot (NWU/NPR) \quad (13.61)$
Winning down Periods (%)	$WDP = 100 \cdot (NWD/NPF) \quad (13.62)$
<i>Source: Dunis et al. (2004)</i>	

SUMMARY

We have introduced high frequency trading and demystified the black box. High frequency trading is an application of systematic and quantitative algorithms for the trading of securities. Critical to an effective strategy is soundly curated high frequency data coupled with a rigorous research framework. We described a number of practical trading strategies alongside examples. These strategies form the alpha model of the black box.

Statistical arbitrage capitalizes on statistical misrepresentations in the prices of related securities. Triangular arbitrage assesses triangular relations between currencies, taking positions when these relations are broken. Liquidity trading attempts to make the spread in the high frequency domain, focusing on the number of winners rather than on the volume per given trade. Market-neutral arbitrage builds on classical relations defined by CAPM to take long-short positions in mispriced securities. Index arbitrage exploits discrepancies in the Law of One Price, taking relative positions to capitalize. And finally, merger arbitrage aims to capitalize on mispricing dynamics prior to deal completion.